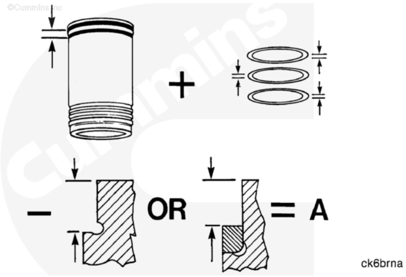
Trainings

Machine



Read the complete procedure before beginning to machine the block.

The counterbore ledge can be machined. Shims will be installed to obtain the correct cylinder liner protrusion.

The cylinder liner protrusion (A) is the total sum of the thickness of the liner flange and the shims, minus the counterbore depth.

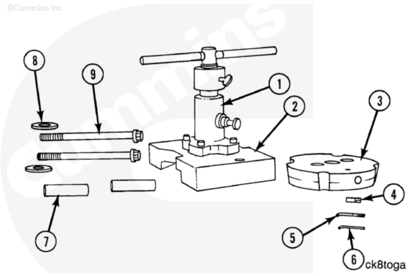
Refer to Procedure 001-064 in Section 1 (/qs3/pubsys2/xml/en/procedures/28/28-001-064-tr.html)

Note : Some liners are oversize in flange thickness and flange outside diameter.

If necessary, machine the block for oversize liners before machining the counterbore ledge for depth.

Refer to Procedure 001-026 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)

Resurface the block as required before machining the counterbore ledge depth. See the Alternative Repair Manual, Bulletin 3379035.



Tools Required:

1. Drive unit, Part Number 3376685
2. Adapter plate, Part Number 3376687
3. Cutter plate, Part Number 3375980
4. Tool bit (Part number depends on block style.)
5. Hex wrench [0.1875 inch]
6. Hex wrench [0.0938 inch]
7. Bolt spacers [0.8125 inch inside diameter x 3.500 inch]
8. Plain washers [0.8125 inch inside diameter x 2.000 inch outside diameter]
9. Cylinder head capscrews for the block being serviced.

Note : The cylinder block counterbore tool, Part Number 3376684, contains the tools listed above. The

same kit contains the tools to machine the counterbore depth on all Cummins® engines except the L10.

Install the drive unit (1) on the adapter plate (2).

Torque Value: 41 n•m [30 ft-lb]

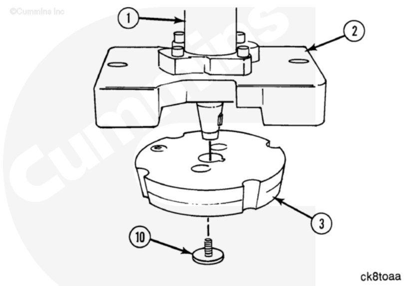
Align the key slot on the cutter plate (3) with the key on the drive unit shaft. Install the plain washer and the capscrew (10).

Torque Value: 41 n•m [30 ft-lb]

Use a mallet. Tap the cutter plate. The cutter plate **must** be seated on the shaft.

Refer to Procedure 001-026 in Section 1 (/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)

Note : Radius specifications are [inch].



Two different tool bits are approved to machine the K38 and K50 cylinder block counterbore ledge depth.

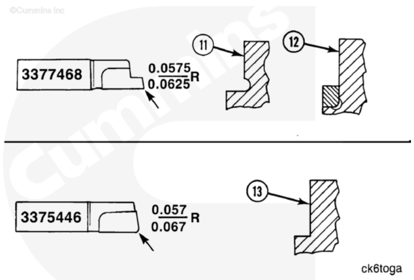
- Tool bit, Part Number 3377468, **must** be used to machine all blocks that contain the double undercut radius.

(11) Thin Deck Block - Double Undercut

(12) Thick Deck Block - Counterbore Ring

Note : The counterbore ring must be removed before machining the block style (12).

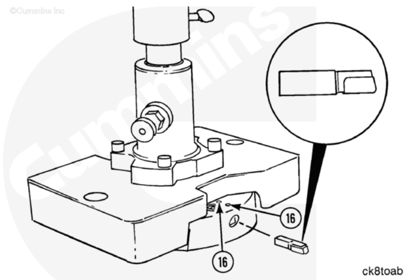
- Tool bit, Part Number 3375446, **must** be used to machine the thin deck design block (13).



Install the correct tool bit in the cutter plate. Position the bit so it cuts when the plate is moving **clockwise**.

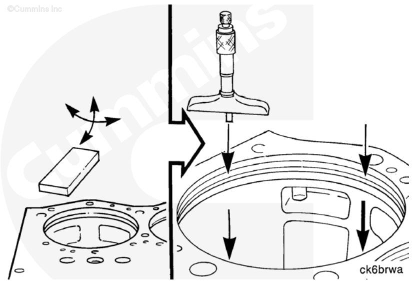
Push the tool bit into the plate until it is even with, or below, the outside diameter of the plate.

Some resistance can be felt when installing the bit extension device. If the bit does **not** go in, loosen the two locking screws (16).



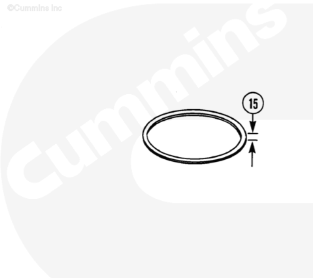
Use a fine hard type stone. Remove any nicks and burrs from the head surface and the counterbore ledge. Nicks and burrs prevent the tool from seating correctly.

Use a depth micrometer to measure the counterbore depth. Measure at four equally spaced locations. Record the measurements.



⚠ CAUTION ⚠

Do not use more than three shims under one liner. The use of one, thick shim is better than two thin shims.

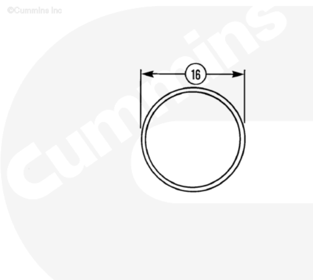


Measure the thickness (15) of the shims for the engine block being machined.

Select the best combination of shims and counterbore rings to obtain the correct protrusion.

Counterbore Shim Thickness (15)		
mm	Block Style	in
0.051	All	0.020
0.079	All	0.031
1.575	All	0.062

Counterbore shims are available with a different outside diameter (16).



Shim outside diameter for all blocks WITH Double Undercut Radius

WITH Double Undercut Radius		
mm		in
189.79	MIN	7.472
190.04	MAX	7.482

Shim outside diameter for all K38 or K50 Blocks WITHOUT Double Undercut Radius

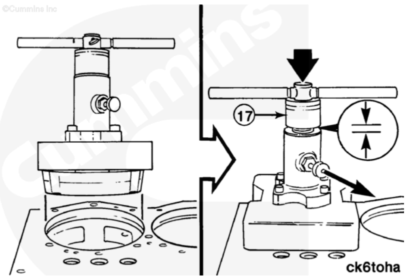
WITHOUT Double Undercut Radius		
mm		in
187.20	MIN	7.370
187.45	MAX	7.380

Note : Counterbore rings that are oversize in thickness help to adjust the liner protrusion.

Counterbore Ring Thickness		
mm	Oversize [Inch]	in
5.055 to 5.067	Standard	0.1990 to 0.1995
5.105 to 5.118	[0.002]	0.2010 to 0.2015
5.156 to 5.169	[0.004]	0.2030 to 0.2035

⚠ CAUTION ⚠

Do not allow the weight of the cutter plate to cause it to fall. The bit plate will be damaged. Pull up on the shaft when pulling the locking pin.



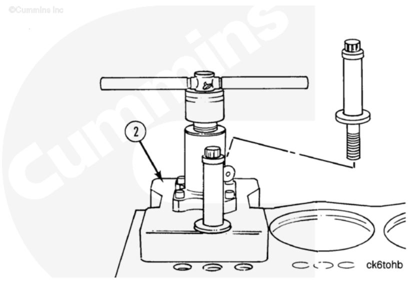
Put the cylinder block counterbore tool on the block.

Pull the locking pin. Lower the cutter plate until it touches the counterbore ledge. The taper on the cutter plate centers the tool in the cylinder bore.

There **must** be clearance between the stop collar (17) and the housing. Turn the collar until there is clearance.

Turn the shaft backward and forward. The taper **must** be positioned squarely in the bore.

There **must** be no clearance between the adapter plate mounting surface and the block head surface.

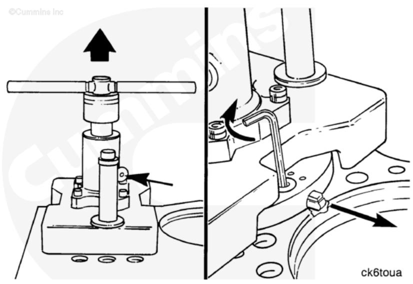


Align the holes in the adapter plate (2) with two cylinder head capscrew holes in the block. Install the plain washers, spacers, and the cylinder head capscrews.

Torque Value: 68 n•m [50 ft-lb]

If the tool is **not** aligned correctly, it does **not** turn after torque is applied to the capscrews.

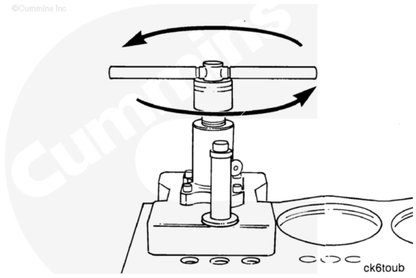
Pull the tool up until the spring loaded locking pin moves in. This will hold the cutter plate up.



Use a [0.1875 inch] hex wrench. Turn the bit extension device until the tool bit touches the counterbore inside diameter.

⚠ CAUTION ⚠

Turn the handle counterclockwise. The tool will damage the bore if it is turned in the opposite direction.

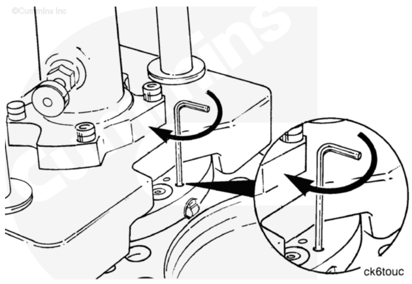


Use the handle to turn the tool **counterclockwise**. Listen for the sound of the tool bit scraping the bore wall as it is turned a minimum of 180 degrees.

If the sound does **not** continue, use the bit extension device to push the tool bit against the bore opposite the first position.

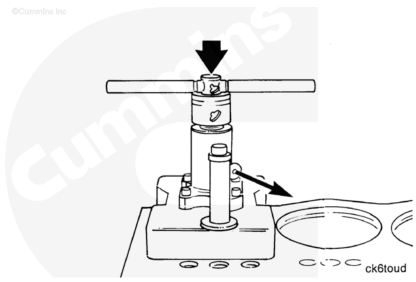
Turn the tool another 180 degrees. Listen again for the sound of the bit. Check for a burr if it touches at one point only.

Use a [0.9375 inch] hex wrench. Tighten the two locking screws of the tool bit.



⚠ CAUTION ⚠

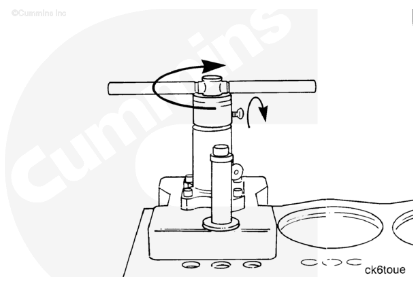
Do not allow the weight of the cutter plate to cause it to fall. The tool bit will be damaged. Pull up on the shaft when pulling the locking pin.



Pull the locking pin. Lower the tool so that the tool bit touches the counterbore ledge.

Note : If the block has the large double undercut radius, proceed to the next illustration. If it does not have the undercut radius, proceed five illustrations to adjust the depth of machining.

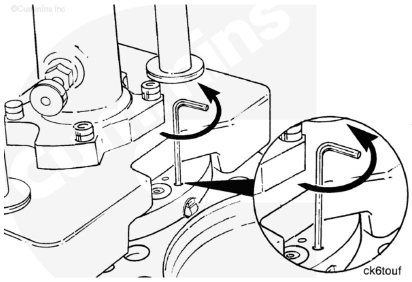
On blocks with a double undercut radius, the tool bit **must** be adjusted so it is in the radius.



Use the stop collar to raise the tool bit from the counterbore ledge 0.025 mm to 0.25 mm [0.001 inch to 0.010 inch].

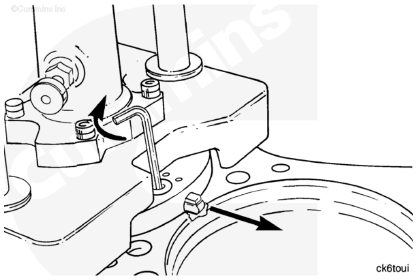
The tool **must** be raised enough to allow the bit extension device to function.

Use a [0.9375 inch] hex wrench. Loosen the tool bit locking screws **only** enough to allow the extension device to function.



Use the [0.1875 inch] hex wrench. Turn the bit extension device until the tool bit touches the inside diameter of the large radius.

This **must** be done to make sure the large radius is extended downward as the ledge is machined.



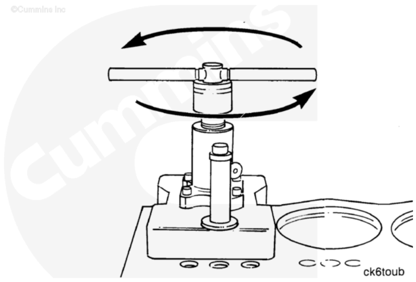
⚠ CAUTION ⚠

Turn the handle counterclockwise. The tool will damage the bore if it is turned the opposite direction.

Use the handle to turn the tool **counterclockwise**. Listen for the sound of the tool bit scraping the bore wall as it is turned a minimum of 180 degrees.

If the sound does **not** continue, use the bit extension device to push the tool bit against the bore wall opposite the first position.

Turn the tool another 180 degrees. Listen again for the sound of the bit. Check for a burr if it touches at one point **only**.

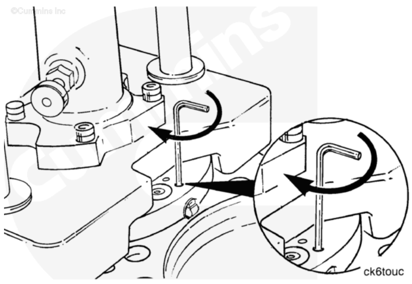


⚠ CAUTION ⚠

Do not attempt to raise the cutter plate with the tool bit extended into the double undercut radius. Damaged can result.

Use a [0.9375 inch] hex wrench. Tighten the two locking screws of the tool bit.

Turn the stop collar to lower the cutter plate so that the bit touches the ledge.



Adjust the Depth of Machining

Make sure the tool bit is seated on the counterbore ledge.

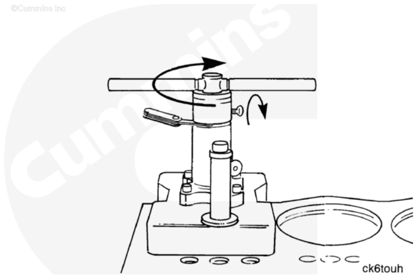
Results will be correct with several 0.05 mm to 0.08 mm [0.002 inch to 0.003 inch] cuts. The tool bit will be damaged by deep cuts. The surface finish will be excessively rough.

Note : Surface finish specification must be 80AA or better.

Make sure the tool bit is **not** raised.

Turn the stop collar **clockwise** to adjust the clearance.

Move a feeler gauge around the housing. Clearance **must** be within specifications at the nearest point.



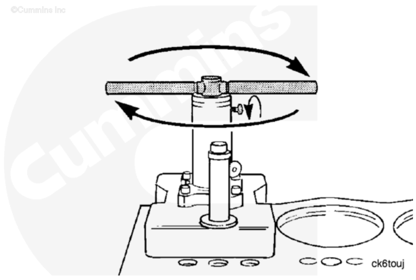
Tighten the thumb screw on the stop collar.

Note : Record the depth of each machining. The figures are required to obtain the correct liner protrusion.

Apply downward pressure. Turn the handle **clockwise**. Keep the tool bit against the ledge.

Machine the bore until the stop collar contacts the housing. **Not** using pressure, spin the tool two or three times to smooth the surface.

Machine the counterbore in several operations until the maximum liner protrusion is obtained.



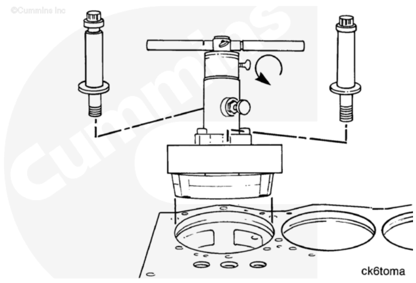
⚠ CAUTION ⚠

Loosen and retract the tool bit before attempting to raise the tool.

Raise the cutter plate.

Remove the tool from the block.

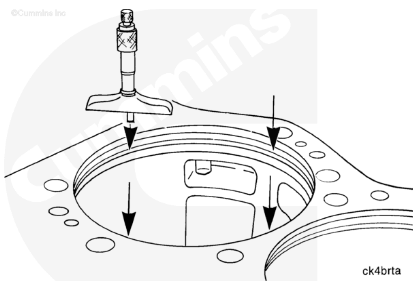
Use a fine hard type stone. Remove any burrs and sharp edges from the counterbore ledge.



⚠ CAUTION ⚠

The micrometer must not touch the radius on a block that does not contain a double undercut counterbore radius.

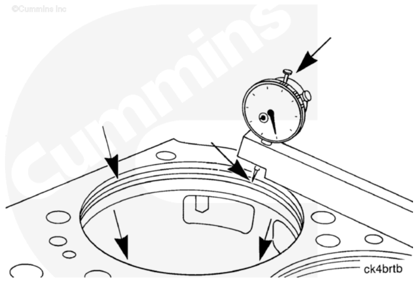
Use a depth micrometer. Measure the counterbore depth in the four locations illustrated.



The four measurements **must not** vary more than 0.25 mm [0.01 inch]. If the measurements exceed the specification, the counterbore ledge **must** be machined again.

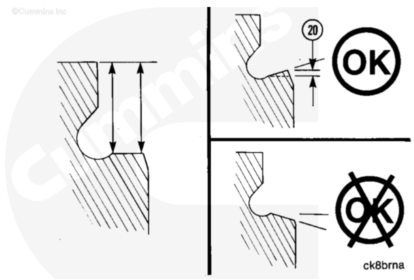
⚠ CAUTION ⚠

The indicator must not touch the counterbore radius on a block that does not have a double undercut.



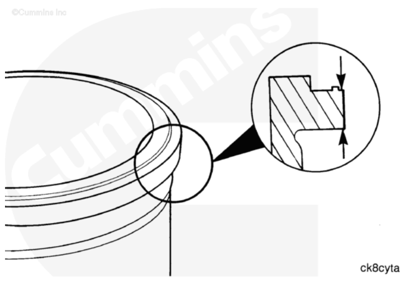
Use a gauge block, Part Number 3376220. Measure the angle of the counterbore ledge at four equally spaced locations.

The measurement of the ledge depth **must** be performed as near to the counterbore radius as possible, and as near to the counterbore edge as possible.



The angle (12) is OK if the measurement that is near the counterbore edge is the same or no more than 0.036 mm [0.0014 inch] shorter than the measurement near the counterbore radius.

Machine the ledge again if the measurement near the counterbore edge is longer than the measurement near the radius.

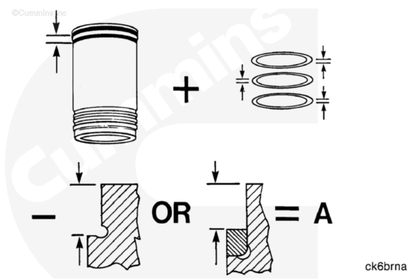


Check the counterbore for cracks.

Refer to Procedure 001-026 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)

Note : Liners and shims must be identified by the cylinder number for proper assembly.

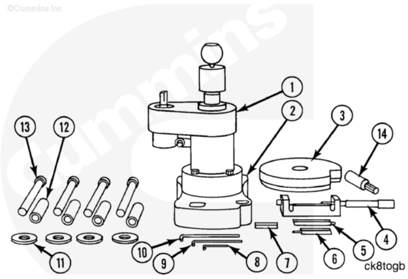
Measure the liner flange thickness at four equally spaced locations around the flange.



Cylinder liner protrusion (A) is the total sum of the liner flange thickness and shim thickness, minus the depth of the counterbore.

Cylinder Liner Protrusion (A)		
mm		in
0.15	MIN	0.006
0.20	MAX	0.008

Tools required for this procedure are listed below. They are available separately or in kits.



Liner Counterbore Tool Kit, Part Number 3377356.

- 1. Liner Counterbore Drive Unit, Part Number 3823352
- 8. [0.9375 inch] Hex Key, Part Number ST-1168-57, or equivalent
- 9. [0.1563 inch] Hex Key, Part Number ST-1168-58, or equivalent
- 10. [0.1875 inch] Hex Key, Part Number ST-1168-60, or equivalent.

Liner Counterbore Salvage Tool Kit, Part Number 3375820

- 2. Base Plate, Part Number 3375827
- 3. Cutter Plate Assembly, Part Number 3824054
- 4. Micrometer Assembly, Part Number 3375826
- 5. Setting Standard, Part Number 3375828
- 6. Cutter Assembly (Part Number depends on block type.) See following text
- 7. Depth Spacer Block (Part Number depends on block type.) See following text
- 11. Plain washers, four, Part Number 3375830, or equivalent. [0.1875 inch inside diameter], [2.000 inch outside diameter] Standard Adapters, Part Number 3375829, or equivalent. [0.8125 inch inside diameter, 3.500 inch length].

The following is also required.

Four cylinder head capscrews, Drill Motor (10-ampere, 450 rpm, 19 mm [0.750 inch] Chuck)

⚠ CAUTION ⚠

Do not attempt to machine blocks of this design for a counterbore ring as it will cause the block to break.

The terms referenced in Procedure 001-026 in Section 1 **must** be understood.

Note : The two types of blocks listed require the counterbore depth to be machined after the upper counterbore inside diameter is machined.

- Counterbore Ring - Thick Deck Design (K38 and K50)
- Thin Deck Block - Double Undercut Radius Design (K38 Only).

These instructions cover the counterbore depth and the upper counterbore inside diameter machining at the same time for the following block.

- Thin Deck Design (K38 **only**)

Note : Cummins Inc. recommends that the accuracy of the tool be checked before machining the customer's block. If possible, machine a scrap block and check the accuracy of the cut.

If a scrap block is **not** available, adjust the tool to machine a size smaller than required. Check the accuracy of the cut. Perform final adjustments before machining to specification.

Note : The cylinder liner inside diameter is the same on oversize liners and standard liners.

Note : The upper counterbore inside diameter must be no more than 0.025 mm [0.001 inch] larger, and no more than 0.076 mm [0.003 inch] smaller than the outside diameter of the cylinder liner on all K38 an K50 blocks.

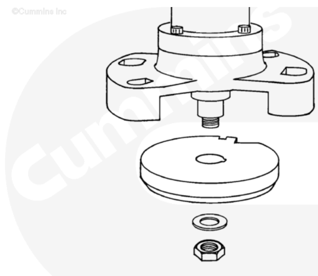
Install the base plate to the main housing. Install the four capscrews.

Torque Value: 41 n•m [30 ft-lb]

Position the cutter plate assembly with the tapered edge away from the housing. Align the keyway in the plate with the key in the shaft.

Install the cutter plate. Install the washer and nut.

Torque Value: 41 n•m [30 ft-lb]

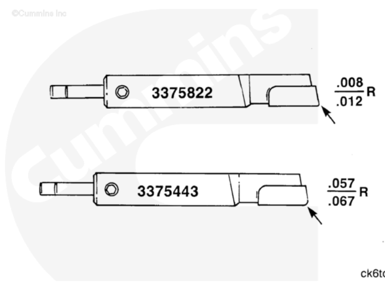


ck8toac

⚠ CAUTION ⚠

Damage to the cylinder block or interference with the cylinder liner will result if the wrong cutter assembly is used.

The following illustrations show the two cutter assemblies that are required with each style of K38 and K50 cylinder blocks.



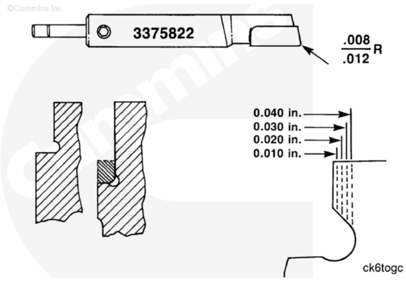
ck6togb

The counterbore ring **must** be removed prior to machining the block. Do **not** machine the ring.

Use cutter assembly, Part Number 3375822, on:

- Thick Deck Blocks - Counterbore Ring
- Thin Deck Blocks - Double Undercut

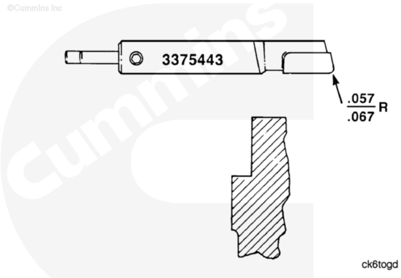
K50 engines contain **only** the thick deck counterbore ring design.



Use cutter assembly, Part Number 3375443, on:

- Thin Deck Blocks

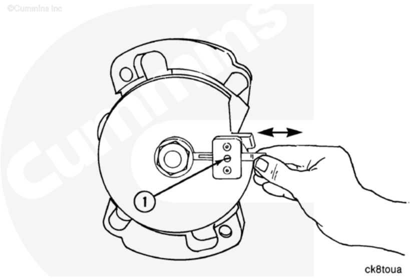
Note : No K50 engines contain counterbores of this design.



Adjust the Set Screw

Put the correct cutter in the cutter plate assembly. Tighten the set screw until it is difficult to slide the cutter in the groove.

Remove the cutter assembly.



Measure the counterbore inside diameter and depth.

Refer to Procedure 001-026 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)

⚠ CAUTION ⚠

The top of the cylinder block must be clean and free of burrs. The tool must be level on the block surface to machine the counterbore inside diameter correctly.

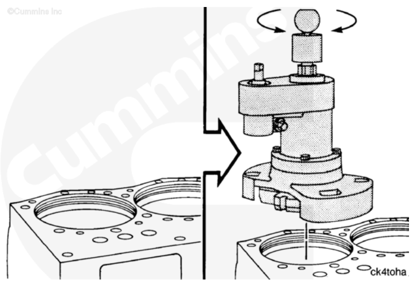
Install the Tool

To "LOCK" the feed mechanism, turn the knob on top **clockwise**.

To "UNLOCK" the feed mechanism, turn the knob **counterclockwise**.

"LOCK" the feed mechanism.

Put the tool on the block.



"UNLOCK" the feed mechanism.

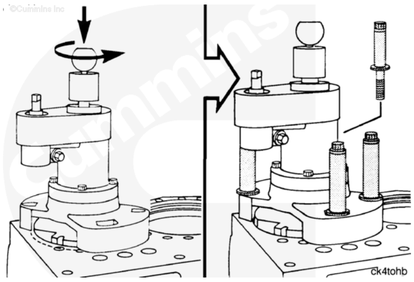
Lower the cutter plate until it touches the lower counterbore inside diameter This will center the tool in the bore.

Make sure the base plate is flat against the block.

Rotate the base plate until the bolt holes align with the cylinder head mounting capscrew locations.

Install the four capscrews, plain washers, and standard adapters.

Torque Value: 47 n•m [35 ft-lb]



Adjust the Cutter Assembly

Use the setting standard, Part Number 3375828. Use the micrometer assembly, Part Number 3375826.

Put the setting standard in the micrometer as illustrated. Measure the length of the standard. In this example, the micrometer indicates [7.600 in].

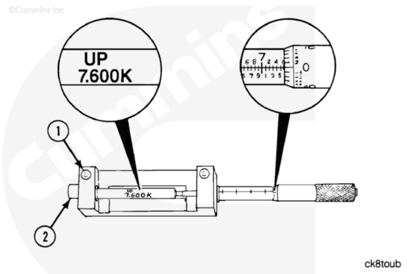
If the reading does **not** agree with the value stamped on the setting standard:

- loosen the set screw (1)
- move the end post (2)
- tighten the set screw (1)

Use one of the next five illustrations to determine the cutter assembly setting.

The dimension listed in Column A is the cutter assembly setting.

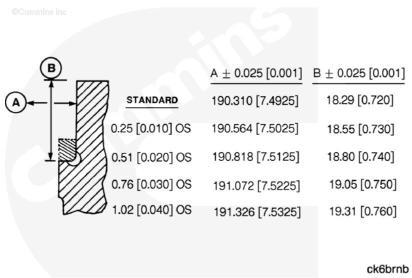
The dimension listed in Column B indicates the size of the depth spacer block that is required.



Note : The counterbore ring must be removed before machining the block.

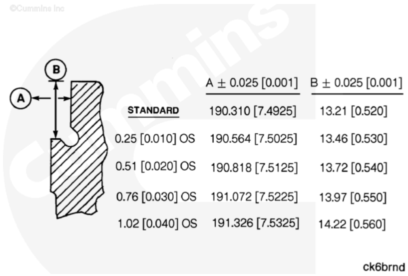
Use cutter assembly, Part Number 3375822, for:

- Thick Deck Blocks - Counterbore Ring



⚠ CAUTION ⚠

Do not attempt to machine blocks of this design for a counterbore ring as it will cause the block to break.

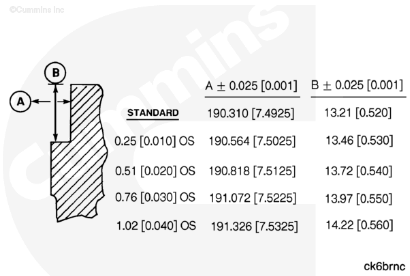


Use cutter assembly, Part Number 3375822, for:

- Thin Deck Blocks Double Undercut

⚠ CAUTION ⚠

Do not attempt to machine blocks of this design for a counterbore ring as it will cause the block to break.



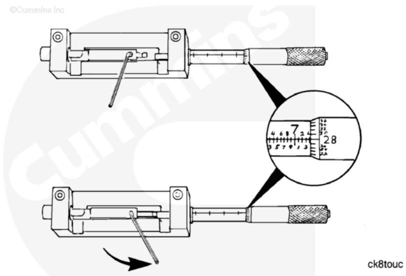
Use cutter assembly, Part Number 3375443, for:

- Thin deck design blocks.

Set the micrometer to the correct dimension for the block being machined.

Put the cutter assembly in the micrometer as illustrated.

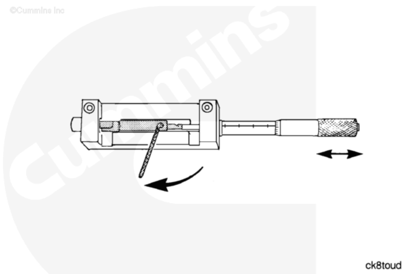
Use a hex key. Loosen the set screw on the cutter assembly. Move the spring loaded plunger until it contacts the micrometer.



Tighten the set screw.

Move the micrometer to check the setting.

Repeat the setting process until the cutter assembly is set accurately.

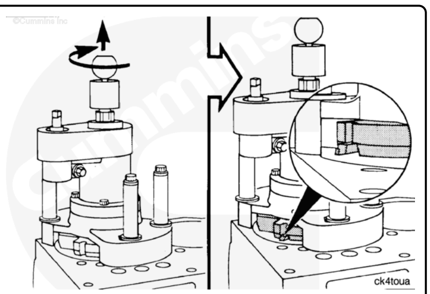


Adjust Machining Depth

Raise and "LOCK" the drive assembly.

Put the cutter assembly in the cutter plate. It **must** extend a minimum of 6 mm [0.250 inch] over the edge of the counterbore.

Tighten the top set screw for the cutter bit.

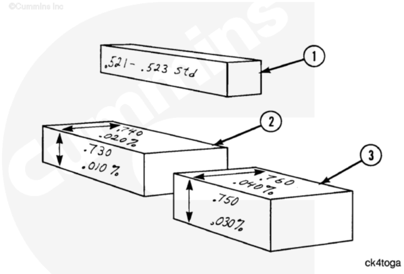


Choose the correct depth spacer block.

See Column B in the illustration specifying machining dimensions for the style of block being machined. Spacer block thickness **must** match the value in Column B.

- Part Number 3376189
- Part Number 3375824
- Part Number 3375831.

These depth spacer blocks are contained in the liner counterbore conversion kit, Part Number 3375820.



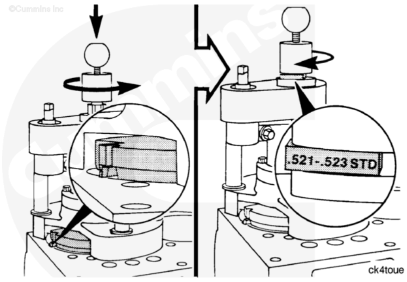
Note : The spacer blocks, Part Numbers 3375824 and 3375831, are for the thick deck counterbore ring only. Part Number 3376189, is for all other blocks.

"UNLOCK" the drive mechanism. Lower the cutter plate until the cutter touches the block.

Put the correct depth spacer block on the tool. The depth of the cut dimension **must** be visible (as illustrated).

Turn the stop collar until it touches the depth spacer block as illustrated.

Note : The 13.3 mm [0.522 in] depth spacer block is illustrated as an example only.



Final Adjustments for Depth

The style of the block being machined determines the next adjustment.

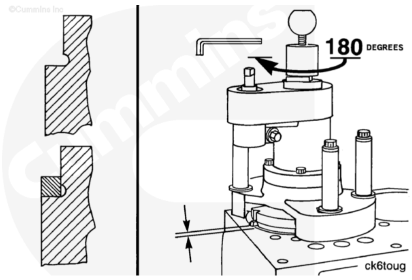
See the following illustrations. Follow the directions given for the illustration that applies to the block being repaired.

Follow these instructions for the

- Thick Deck Block - Counterbore Ring
- Thin Deck Block - Double Undercut

⚠ CAUTION ⚠

Do not cut the counterbore depth with this tool.



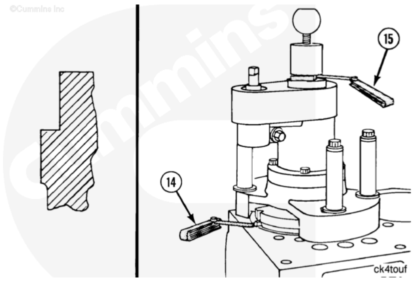
Turn the depth stop collar ½ of a revolution (180 degrees) **counterclockwise**. The cutting assembly **must** be raised slightly from the counterbore ledge.

Use a hex key. Tighten the set screw on the collar.

Follow these instructions for the

- Thin Deck Block

Note : No K50 engines contain this counterbore design.



Determine the difference in depth between the depth required for the oversize being used and the thickness of the spacer block, Part Number 3376189, 13.2 to 13.3 mm [0.521 to 0.523 in] standard.

Loosen the stop collar.

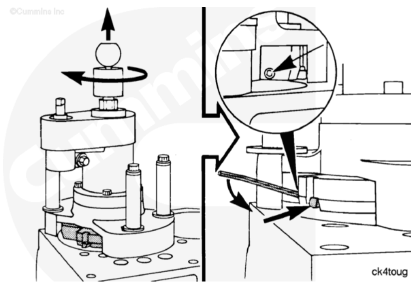
Insert feeler gauge(s) (15) equal to the difference between the spacer block and the stop collar. Turn the stop collar until it touches the feeler gauge. Tighten the set screw on the stop collar. Use a 0.04 mm [0.0015 inch] thick feeler gauge (14) to make sure the tool bit is touching the block.

Note : If the counterbore depth is already greater than the depth required by the oversize, the counterbore depth must be cut to the existing length. Determine the difference between the existing depth and the oversize depth. Insert feeler gauge(s) equal to this difference between the spacer block and the stop collar. Turn the stop collar until it touches the feeler gauge(s). Tighten the set screw on the stop collar. Use a 0.04 mm [0.0015 inch] thick feeler gauge to make sure the tool bit is still touching the block.

Use a hex key. Tighten the set screw on the stop collar.

⚠ CAUTION ⚠

Make sure no dirt or chips are in the cutting assembly groove of the cutter plate.



Machine the counterbore inside diameter

Raise the cutting plate assembly. "LOCK" the drive mechanism.

Push the cutting assembly into the cutter plate until it touches the shaft.

Use a hex key. Tighten the set screw.

⚠ WARNING ⚠

Hold the drill firmly. The drill will be difficult to hold when the cutter bit initially touches the block.

Lower the cutting plate until the cutting assembly is 1.59 mm [0.0625 in] above the block. "LOCK" the drive mechanism.

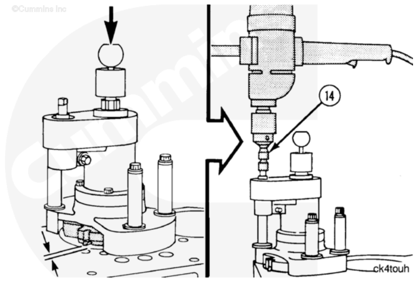
Install the universal drive (14) in the drill.

Note : The feed mechanism automatically controls the rate that the cutter is lowered.

"START" the drill. The cutter plate will free-wheel after the depth stop collar has contacted the drive unit.

"STOP" the drill after the cutter has free-wheeled for five to ten revolutions.

Remove the drill.

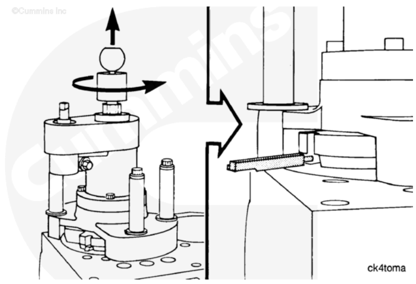


"UNLOCK" the drive mechanism.

Raise the cutter plate.

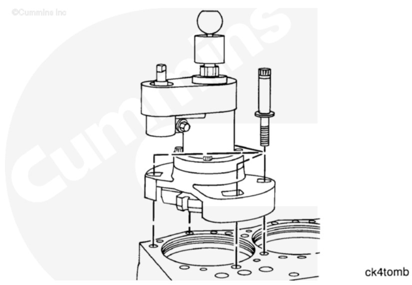
"LOCK" the drive mechanism.

Use a hex key. Loosen the set screw. Remove the cutter assembly.



Remove the mounting capscrews, adapters, and washers.

Remove the tool.

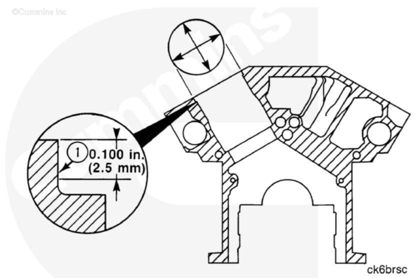


Use a hone stone. Remove burrs from the sharp corner of the inside diameter.

Measure the upper counterbore inside diameter (1) in the location shown.

Machine the diameter again if the inside diameter is smaller than specification.

Machine the block for the next larger size liner when the inside diameter is larger than specification.

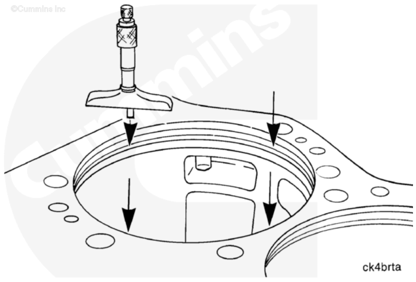


Apply the next three steps to the thin deck block style **only**.

For the counterbore ring - thick deck and thin deck double-undercut radius designs, follow the instructions to cut the counterbore depth to the oversize specifications.

⚠ CAUTION ⚠

The micrometer must not contact the counterbore radius on a block that does not contain a double undercut radius.

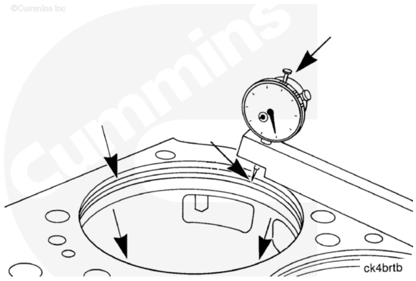


Use a depth micrometer to measure the counterbore depth in the four places illustrated.

The four measurements **must not** vary more than 0.25 mm [0.001 in]. The counterbore ledge **must** be machined when the measurements vary.

⚠ CAUTION ⚠

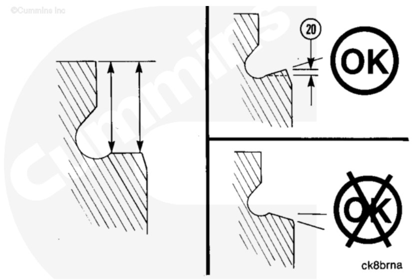
The indicator must not contact the counterbore radius on a block that does not contain a double undercut radius.



Use a gauge block, Part Number 3376220, to measure the angle of the counterbore ledge at four-equally spaced locations around the counterbore.

The measurement of the ledge depth **must** be performed as near to the counterbore radius as possible, and as near to the counterbore edge as possible.

The angle (20) of the counterbore ledge is OK if the measurement that is near the counterbore edge is the same or no more than 0.036 mm [0.0014 in] shorter than the measurement near the counterbore radius.



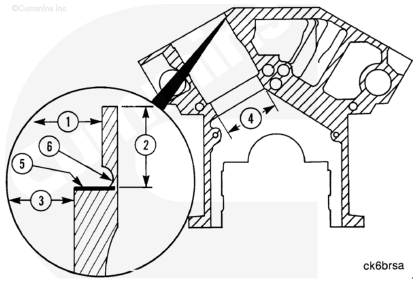
Machine the ledge again when the measurement near the counterbore edge is longer than the measurement near the counterbore radius.

Check the concentricity between the upper counterbore inside diameter (1) and the packing ring bore (4).

Refer to Procedure 001-070 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-070-tr.html)

Clean the cylinder block.

Refer to Procedure 001-026 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html)



Repair

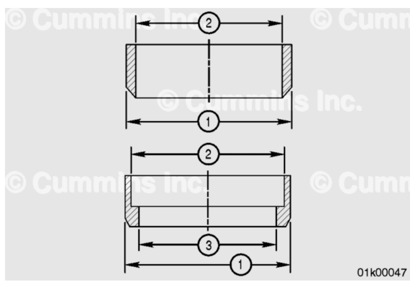
For machining the bores, use a capable machine shop.

Select the correct repair sleeve from the upper deck repair bushings/tooling, Part Number 3822959. Reference Section 11 of the Service Products Catalog, Bulletin 3377710.

Measure and record the outer diameter of the repair sleeve.

Note : The inner diameters of the repair sleeves are ready-machined to the nominal counter bores diameters.

Repair Sleeve Counterbore Dimensions when Installed in Block Bored at Reference Diameter		
Standard Size Repair Sleeve		
	Minimum	**Maximum**
	mm [in]	**mm [in]**
Cylinder Block Reference Diameter - Upper Counterbore	193.015 [7.5990]	193.040 [7.6000]



Repair Sleeve Counterbore Dimensions when Installed in Block Bored at Reference Diameter		
Standard Size Repair Sleeve		
	Minimum	**Maximum**
	mm [in]	**mm [in]**
Straight Sleeve - Outer Diameter (1) Before Installation in Block	193.142 [7.6040]	193.167 [7.6050]
Straight Sleeve - Outer Diameter (1)	193.015 [7.5990]	193.040 [7.6000]
Straight Sleeve - Upper Counterbore (2)	190.297 [7.4920]	190.297 [7.4930]
Cylinder Block Reference Diameter - Upper Counterbore	193.269 [7.6090]	193.294 [7.6100]
“L” Shaped Sleeve - Outer Diameter (1) Before Installation in Block	193.396 [7.6140]	193.421 [7.6150]
“L” Shaped Sleeve - Outer Diameter (1)	193.269 [7.6090]	193.294 [7.6100]
“L” Shaped Sleeve - Upper Counterbore (2)	190.297 [7.4920]	190.297 [7.4930]
“L” Shaped Sleeve - Lower Counterbore (3)	181.788 [7.1570]	181.813 [7.1580]

2.54 mm [0.100 in] Oversize Repair Sleeve		
	Minimum	**Maximum**
	mm [in]	**mm [in]**
Cylinder Block Reference Diameter - Upper Counterbore	195.555 [7.6990]	195.580 [7.7000]

2.54 mm [0.100 in] Oversize Repair Sleeve		
	Minimum	**Maximum**
	mm [in]	**mm [in]**
Straight Sleeve - Outer Diameter (1) Before Installation in Block	195.682 [7.7040]	195.707 [7.7050]
Straight Sleeve - Outer Diameter (1)	195.555 [7.6990]	195.580 [7.7000]
Straight Sleeve - Upper Counterbore (2)	190.297 [7.4920]	190.297 [7.4930]
Cylinder Block Reference Diameter - Upper Counterbore	195.809 [7.7090]	195.834 [7.7100]
Straight Sleeve - Outer Diameter (1) Before Installation in Block	195.936 [7.7140]	195.961 [7.7150]
“L” Shaped Sleeve - Outer Diameter (1)	195.809 [7.7090]	195.834 [7.1000]
“L” Shaped Sleeve - Upper Counterbore (2)	190.297 [7.4920]	190.297 [7.4930]
“L” Shaped Sleeve - Lower Counterbore (3)	181.788 [7.1570]	181.813 [7.1580]

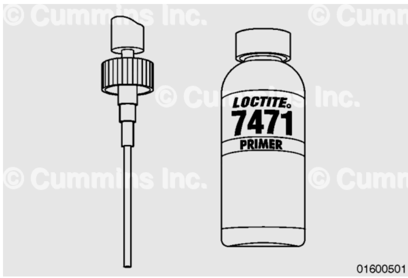
Machine the top liner bore to give interference and depth according to the table below.

	Interference Fit on Diameter	Depth	Corner Radius
	Min/Max	Min/Max	Min/Max
	mm [in]	mm [in]	mm [in]
Straight Sleeve	0.102 - 0.152[0.004 - 0.006]	12.50 - 12.75[0.492 - 0.502]	0.25 - 0.50[0.010 - 0.020]
“L” Shaped Sleeve	0.102 - 0.152[0.004 - 0.006]	21.33 - 21.46[0.840 - 0.845]	1.78 - 2.03[0.07 - 0.08]

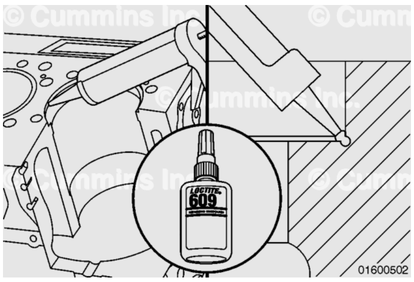
⚠ WARNING ⚠

Use skin and eye protection when handling caustic solutions to reduce the possibility of personal injury.

Thoroughly clean the outer diameter of the repair sleeve and cylinder block bore. Use cleaner/primer, Part Number 3824715.

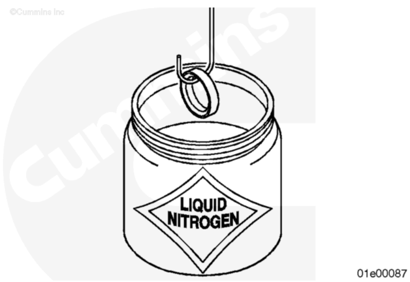


Apply retaining compound, Part Number 3823718 or equivalent, to the corner of the cylinder block counterbore.



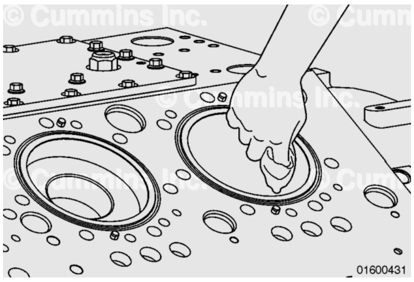
Use liquid nitrogen to freeze the repair sleeve.

Install the sleeve into the cylinder block. Make sure that the repair sleeve bottoms in the counterbore.



Remove all traces of surplus retaining compound from the cylinder block.

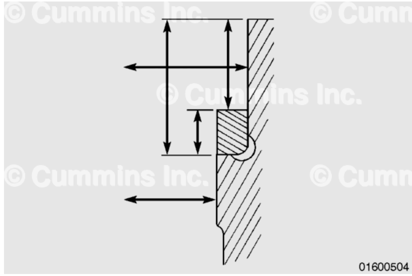
Allow three to four hours for the retaining compound to cure.



Machine the repair sleeve to the present specifications. Refer to Procedure 001-027 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-027.html)

Replace the undercut with a corner radius.

Corner Radius Specification		
mm		in
1.270	MIN	0.050
1.527	MAX	0.060



Clean and inspect the cylinder block.

Refer to Procedure 001-027 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-027.html)

Last Modified: 29-Jun-2021